Statistical Methods in Education Research
Vanderbilt University
Prof. Sean P. Corcoran

Due November 3, 2026
63 total points

Problem Set 8

Instructions: Answer the following questions in their entirety in a separate document. Submit your completed problem set as a PDF via email to `sean.corcoran@vanderbilt.edu`. Use your last name and problem set number as the filename. Working together is encouraged, but it is expected that all submitted work be that of the individual student.

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1. Data are gathered on 16 students. Half of the students are randomly assigned to a new tutoring program and half have their usual schooling experience. A study finds that test scores for the tutoring program students are 10 points higher on average than those for the other students (p=*0.3*). The mean test score overall is 200 with a standard deviation of 40. (**3 points—1 each**)

   (a) Is the study's finding statistically significant? Explain why or why not.

   (b) Is the study's finding practically significant, in your opinion? Explain what practical significance means here.

   (c) Would you conclude that the tutoring program is effective based on these results? Ineffective? Briefly explain.

2. Two studies were commissioned to evaluate an intensive program designed to enhance social and emotional learning (SEL) among adolescents. The index used to measure SEL has a mean of 50 and a standard deviation of 10. Study 1 failed to find a statistically significant improvement in SEL, with a 95% confidence interval for the gain in SEL of $(-7, 17)$. Study 2 also failed to find a statistically significant improvement in SEL, with a 95% confidence interval of $(-2.5, 1.5)$. Which of these two studies (if any) is more informative to a policymaker, in your opinion, and why? (**4 points**)

3. A decision is planned in a test of $H_0 : \mu = 0$ against the alternative $H_a : \mu > 0$, using $\alpha = 0.05$. If the actual μ is 5, the probability of a Type II error (β) is 0.17. (**6 points—2 each**)

   (a) Explain the meaning of the last sentence in words.

   (b) If the test used $\alpha = 0.01$, would the probability of a Type II error be less than, equal to, or greater than 0.17? Explain.

   (c) If μ were actually 10, would the probability of a Type II error be less than, equal to, or greater than 0.17? Explain.

4. Your team is tasked with studying whether charter school students perform *better* in mathematics, on average, than traditional public school students. You know that math scores in the population of public school students have a normal distribution with a mean of 420 and standard deviation of 84. You plan to administer the same test to a sample of charter school students to test your hypothesis. Use Stata or a web applet to answer the following questions. (**14 points**)

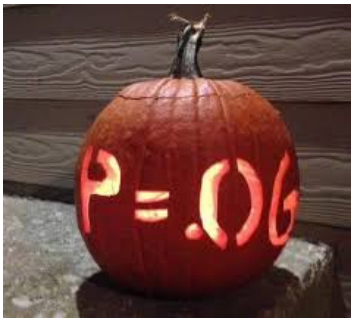
   (a) What are the null and alternative hypotheses in this problem? (**2 points**)

   (b) Describe what a Type I and Type II error would be in this study. (**2 points**)

   (c) You consider a meaningful difference in test scores to be +21 points. What specific alternative μ_a would this imply? How large an *effect size* (Cohen's d) would this be? (**4 points**)

   (d) If you were to randomly sample 40 charter school students to take the math test, what is the power associated with your significance test (assuming the alternative in part c)? What is the probability of a Type II error? Use significance level $\alpha = 0.05$. (**3 points**)

   (e) Holding everything else constant, how large of a sample would you need in order to increase the power to 0.80? (**3 points**)



5. Your favorite candidate currently has a favorability rating of 49%. You plan to take a random sample of 100 likely voters to assess whether her favorability has *improved* (i.e., is greater than 49%). Use Stata to answer the following questions, and use $\alpha = 0.05$. (Hint: you can use the power calculator for proportions called `power oneproportion`): (**8 points**)

   (a) Write down the null and alternative hypotheses for this test. (**2 points**)

   (b) What is the statistical power of your test if the true favorability rating is 55%? Provide an explanation in words. (**2 points**)

(c) Have Stata produce a table and graph of power for alternative hypotheses ranging from 50% to 65%, in increments of 5%. What is the smallest alternative that can be detected with 80% power? (**4 points**)

6. A recent paper by von Hippel (2025) argues that effect sizes expressed in standard deviation units are not easily understood by educators (see also Lortie-Forgues, Sio, and Inglis 2021 on this point). He recommends that these be converted to more relatable units, such as percentile points. A challenge with this is the percentile point change implied by a delta sd effect depends on the starting point. von Hippel shows that–for a wide range of values–you can get a pretty good approximation of the percentile point change by multiplying the effect size in sd units by 37. Explore these ideas using simulated data in Stata following the steps below. (**10 points**)

   - Create an empty dataset with 1000 observations.
   - Create a new variable $x1$ that contains random draws from the standard normal distribution. We will assume the outcome of interest (e.g., a test score) follows this distribution.
   - Create a new variable $x1p$ that contains the percentile of the standard normal distribution that each value of $x1$ represents (Hint: how can we find the percent of all values in a standard normal that fall below a particular value $x1$)?
   - Suppose an educational intervention has a 0.25 sd effect on all student scores. Create a new variable $x2$ that equals $x1 + 0.25$. According to the von Hippel rule of thumb, what is the approximate effect in percentage point terms?
   - Create a new variable $x2p$ that contains the percentile rank of the standard normal distribution that each value of $x2$ represents and then a new variable *chgp* that contans the change in percentile rank ($x2p - x1p$).
   - Create a scatter plot (`scatter y x`) that plots the *chgp* against the starting score $x1p$. What do you observe?
   - Consider the starting points $x1p$ in the interval [0.2, 0.8]—i.e. the middle 3/5 of the distribution. Summarize the *chgp* in this interval. How closely does it correspond to the von Hippel rule of thumb? (You may also see this in the scatter).

7. The Stata syntax below will create a dataset with 200 sample means for $\bar{x}_1$ and $\bar{x}_2$, and 200 sample standard deviations s_1 and s_2. Each sample consists of $n = 50$ random draws from normal distributions in which $x_1 \sim N(0, 5)$ and $x_2 \sim N(2, 5)$. (That is, x_1 has a normal distribution with mean zero and standard deviation 5; x_2 has a normal distribution with mean 2 and standard deviation 5). (**18 points**)

```
set seed #

forvalues j=1/200 {
   clear
   tempfile sample‘j’
   set obs 50
   gen x1 = rnormal(0,5)
   gen x2 = rnormal(2,5)
   collapse (mean) x1 x2 (sd) sx1=x1 sx2=x2
   gen n= 50
   save ‘sample‘j’’
   }
use ‘sample1’, clear
forvalues j=2/200 {
   append using ‘sample‘j’’
   }
```

(a) Run the Stata syntax above, and choose your own starting seed value (#). Take care that your quotations are correct, as they may not copy over correctly from this document into a do-file. Use the `twoway` graphing function with `kdensity` to overlay the resulting distributions of $\bar{x}_1$ and $\bar{x}_2$ (see below). `kdensity` is a kernel density function, a kind of "smoothed" histogram. (**3 points**)

```
graph twoway (kdensity x1) (kdensity x2)
```

(b) Create two new variables that contain the lower and upper bounds of a 95% confidence interval for μ based on $\bar{x}_1$. (Each resulting $\bar{x}_1$—all 200 of them—has a confidence interval associated with it). Be sure to use the *sample* standard deviation when creating your confidence intervals, not the population standard deviation, and continue to do so in part (c). In what percentage of samples does your confidence interval contain the true population mean $\mu = 0$? In what percentage does it not? (Hint: you can create a third variable that flags whether the CI contains zero or doesn't). (**3 points**)

(c) Create two new variables that equal the t-statistics and p-values from a test of $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$. (Each resulting $\bar{x}_1$ will have a t and p-value for this test). Note this is a two-sided test. Create a histogram to show the resulting distribution of p-values across the 200 random samples. In what percentage of your random samples is $p < 0.05$? Explain in words what this percentage represents. (**4 points**)

(d) You are still interested in the null hypothesis $H_0 : \mu = 0$. However, suppose that—unbeknownst to you—the sample means $\bar{x}_2$ (not the $\bar{x}_1$ that you used in

parts b-c) reflect draws from the true distribution of x. That is, x has a mean of 2, not 0. As in part (b), create two new variables that contain the lower and upper bounds of a 95% confidence interval for μ, based on $\bar{x}_2$ (and s_2). In what percentage of random samples does your confidence interval based on $\bar{x}_2$ contain *zero*? Explain in words what this percentage represents. (**4 points**)

(e) Repeat part (c) using $\bar{x}_2$ and continue to test the hypothesis $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$. In what percentage of these random samples is $p < 0.05$? In what percentage of these random samples is $p > 0.05$? Explain in words what these percentages represent. (**4 points**)